



MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Register Description

Table 23: SOC1 Registers (continued)

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x321E	15:0	0x0010	output_control_status_2 (R/W)
	15:11	0x0000	Reserved
	10	0x0000	gated_pixel_clock_output This bit is used to disable the pixel clock during blanking intervals. 0: Pixel clock is not gated during blanking. 1: Pixel clock is gated during blanking.
	9:0	0x0010	Reserved
	Output control and status 2		
R0x326C	15:0	0x1208	2d_aperture_parameters (R/W)
	15	X	Reserved
	14	0x0000	ap_abs Positive Gain Enable 0: Aperture gain is not forced to be positive. 1: Aperture gain is forced to be positive.
	13:11	0x0002	ap_gain_exp 2D aperture gains exponent
	10:8	0x0002	ap_gain_mant 2D aperture gain
	7:0	0x0008	ap_knee 2D aperture threshold
	2d aperture parameters		
R0x327A	15:0	0x002A	red_offset (R/W)
	Offset subtracted from red pixels. This register contains the value for red pixels subtracted by the IFP from raw pixel values before applying lens shading correction and digital gains.		
R0x327C	15:0	0x002A	green1_offset (R/W)
	Offset subtracted from green1 pixels. This register contains the value for green1 pixels subtracted by the IFP from raw pixel values before applying lens shading correction and digital gains.		
R0x327E	15:0	0x002A	green2_offset (R/W)
	Offset subtracted from green2 pixels. This register contains the value for green2 pixels subtracted by the IFP from raw pixel values before applying lens shading correction and digital gains.		
R0x3280	15:0	0x002A	blue_offset (R/W)
	Offset subtracted from blue pixels. This register contains the value for blue pixels subtracted by the IFP from raw pixel values before applying lens shading correction and digital gains.		
R0x328E	15:0	0x000C	dm_edge_th (R/W)
	Demosaic Edge Threshold. This is the value used in demosaic to determine if the current pixel is on an edge.		



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Table 23: SOC1 Registers (continued)

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x3290	15:0	0x0000	test_pattern (R/W)
	15:7	X	Reserved
	6	0x0000	moto_10 Walking 1s test pattern width This bit determines if the walking 1s test pattern is 8-bit or 10-bit wide. 0: 8-bit test pattern 1: 10-bit test pattern
	5	0x0000	en_moto_tp Walking 1s Test Pattern Enable This bit outputs the walking 1's test pattern on the parallel data bus. It can be 8-bit or 10-bit. To configure output stream for the walking 1s test pattern, see also the Output control and status register (0x321C). 0: Walking 1s test pattern is disabled. 1: Walking 1s test pattern is enabled
	4:0	X	Reserved
	Test pattern		
R0x3330	15:0	0x0000	output_format_test (R/W)
	15:9	X	Reserved
	8	0x0000	bypass_lc Enable shading correction bypass Bypass 8+2, after shading correction. Used for calibrating shading correction. 8+2 bypass (0x3330[6] = 1) needs to be enabled to use this mode. 0: Shading correction bypass disabled 1: Shading correction bypass enabled
	7	0x0000	Reserved
	6	0x0000	bypass_8_2 Enable 8+2 bypass output 0: Normal output 1: 8+2 bypass output
	5:3	0x0000	test_ramp These bits enable a test ramp output. The test ramp is generated based on the pixel position, Y (or G) = Cr (or R) = Cb (or B) = {pixel_position_count[4:0], 3'b0}. The output format is determined by 0x332E. 00: Test ramp disabled 01: Increment by column 10: Increment by row 11: Increment by frame
	2	0x0000	cb_dis This bit disables Cb or B channel 0: Normal 1: Disable Cb channel in YUV mode (B in RGB mode)
	1	0x0000	y_dis This bit disables Y or G channel 0: Normal 1: Disable Y channel in YUV mode (G in RGB mode)
	0	0x0000	cr_dis This bit disables Cr or R channel 0: Normal 1: Disable Cr channel in YUV mode (R in RGB mode)
	Output format test		



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Table 23: SOC1 Registers (continued)

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x337C	15:0	0x0006	yuv_ycbcr_control (R/W)
	15:4	X	Reserved
	3	0x0000	clip This bit is used to adjust the range of YUV output for different applications. 0: Y, U and V are not clipped; Y in 0x000–0x00FF. 1: Clip Y in 0x0010–0x00EB; U and V in 0x0010–0x00F0.
	2	0x0001	add_uv_offset Add 0x0080 to U and V 0: No offset added to U and V. 1: Add 0x0080 to U and V.
	1	0x0001	Reserved
	0	0x0000	normalize This bit is used to normalize the range of YUV output for different applications. 0: Y, U and V are not normalized; Y in 0x0000–0x00FF. 1: Normalize Y in 0x0010–0x00EB; U and V in 0x0010–0x00F0.
	YUV YCBCR control		
R0x33F4	15:0	0x0003	kernel_config (R/W)
	15:4	X	Reserved
	3	0x0000	nr_enable Noise reduction enable 0: Noise reduction algorithm disabled 1: Noise reduction algorithm enabled
	2:1	X	Reserved
	0	0x0001	dc_enable Defect correction enable 0: 2D single pixel defect correction is disabled. 1: 2D single pixel defect correction is enabled.
	Kernel configuration		



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Table 24: SOC2 Registers

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x3400	15:0	0x782E	mipi_control (R/W)
	15:10	0x001E	data_type Data Format 0x1E: YUV422 8-bit 0x22: RGB565 0x21: RGB555 0x20: RGB444 0x2A: RAW8 0x2B: RAW10
	9	0x0000	mipi_en Enables logical operation of MIPI 0: Control and data signals are not transmitted to the MIPI interface. MIPI is not necessarily in reset or standby. 1: Control and data signals are allowed from TX logic to MIPI interface.
	8:6	0x0000	chan_num MIPI channel number Virtual channel number to be inserted in MIPI packet header (b1:0 only).
	5	0x0001	Reserved
	4	0x0000	standby_eof Wait until EOF to react to standby 0: Respond to standby_en at end of current packet/line. 1: Respond to standby_en at end of frame.
	3	RO	reg_frame_sync Frame boundary sync bit Asserted when it is safe to update "frame-synchronized" inputs.
	2	0x0001	cont_mipi_clk Continuous MIPI clock Asserted to transmit a continuous MIPI output clock; else clock is only active when data is being transmitted.
	1	0x0001	standby_en MIPI standby Asserted when the chip is to go into standby. The interface completes the transmission of the current packet/frame (as determined by standby_eof), and (for MIPI) sends an end of frame packet and goes into ultra low-power mode. Clock must continue to be supplied until mipi_stdbdy (0x3402[0]) is asserted.
	0	0x0000	restart_en MIPI restart enable When asserted the interface completes the transmission of the current packet/frame and sends an end of frame packet.
MIPI control			



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Table 24: SOC2 Registers (continued)

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x3402	15:0	0x0011	mipi_status (R/W)
	15:6	X	Reserved
	5	RO	mipi_rdy_for_data MIPI ready to receive data Asserted once the MIPI interface is enabled and has completed its wake-up and initialization cycles.
	4	RO	mipiccp_idle MIPI idle Asserted if MIPI interface is not currently transmitting packet data.
	3:1	X	Reserved
	0	RO	mipi_stdby MIPI in standby Asserted when the MIPI interface has completed the transition to ultra low-power state in response to the assertion of standby_en and remains asserted until standby_en is deasserted.
	MIPI status		



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Table 24: SOC2 Registers (continued)

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x3404	15:0	0x0000	custom_short_pkt (R/W)
	15:14	X	Reserved
	13	RO	custom_short_pkt_rst Reset for custom short packet Asserted once the custom short packet has been transmitted to reset custom_short_pkt_req. Since the request may be asserted from an asynchronous, non-continuous clock domain (wrt clk) this signal will remain asserted until it detects that custom_short_pkt_req has transitioned to "0."
	12	0x0000	custom_short_pkt_frame_sync Frame synch for custom short packet When asserted any requested custom short packet will not be transmitted until after the completion of the current frame.
	11	0x0000	custom_short_pkt_req Custom short packet request Asserted to request the insertion of a short packet in the output datastream. Should be left asserted until reset by custom_short_pkt_rst. Must go to "0" before it can be reasserted to make a subsequent request.
	10:8	0x0000	custom_short_pkt_data_type Custom short packet data type The three LSBs of the data type field of a custom short packet.
	7	0x0000	frame_cnt_en Insert frame counter value in WC field Asserted if the frame counter value (8-bit) is to be inserted in the WC field of a start or end of frame short packet (else WC field = 0). Signal should be static for the duration of the frame and should only toggle between frames; that is when frame_reg_sync is asserted. NOTE: This signal masks the frame counter in the short packet WC field but does not stop the counter internally. To ensure the counter starts at 1, frame_cnt_rst must be asserted and released with frame_cnt_en.
	6	0x0000	frame_cnt_rst Reset the frame count sent in the frame start or end short packets. Asserted to reset the frame count transmitted in the frame start or end short packets. Signal must be asserted for at least one "clk" cycle and should only toggle between frames. That is when frame_reg_sync is asserted.
	5:0	X	Reserved
	MIPI custom short packet		
R0x3408	15:0	0x0200	line_byte_cnt (R/W)
	MIPI line byte count The number of data bytes (not including the packet header or footer) in the line. NOTE: The line_byte_cnt must be programmed with a valid value for the data format, as specified by the MIPI or CCP specs. For 10-bit data this means this value should be a multiple of 5, for 12-bit data a multiple of 3 and for YUV/RGB a multiple of 2.		
R0x340C	15:0	0x0000	custom_short_pkt_wc (R/W)
	The contents of the WC field of a custom short packet value to insert in the WC field of a custom short packet specified by R0x3404		



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Table 24: SOC2 Registers (continued)

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x3580	15:0	0x1300	ae_zone_x (R/W)
	15:8	0x0013	ae_zone_x_width Width of each AE zone in x direction divided by 8, minus 1. $w = (ae_zone_x_width + 1) * 8$.
	7:0	0x0000	ae_zone_x_start Start of AE window divided by 8
	AE zone x		
R0x3582	15:0	0x0E00	ae_zone_y (R/W)
	15:8	0x000E	ae_zone_y_width Width of each AE zone in y direction divided by 8, minus 1. $w = (ae_zone_y_width + 1) * 8$
	7:0	0x0000	ae_zone_y_start Start of AE window divided by 8
	AE zone y		
R0x3584	15:0	0x4B00	ae_window_size_lo (R/W)
	Least significant bits of the size of each AE zone in terms of number of pixels.		
R0x3586	15:0	0x0000	ae_window_size_hi (R/W)
	Most significant bit of the size of each AE zone in terms of number of pixels.		
R0x3642	15:0	0x0204	pga_1 (R/W)
	PGA coefficients 1		
R0x3644	15:0	0x0284	pga_2 (R/W)
	PGA coefficients 2		
R0x364E	15:0	0x0010	pga_3 (R/W)
	PGA coefficients 3		
R0x3650	15:0	0x0000	pga_4 (R/W)
	PGA coefficients 4		
R0x3652	15:0	0x0000	pga_5 (R/W)
	PGA coefficients 5		
R0x3654	15:0	0x0000	pga_6 (R/W)
	PGA coefficients 6		
R0x3656	15:0	0x0000	pga_7 (R/W)
	PGA coefficients 7		
R0x3658	15:0	0x0010	pga_8 (R/W)
	PGA coefficients 8		
R0x365A	15:0	0x0000	pga_9 (R/W)
	PGA coefficients 9		
R0x365C	15:0	0x0000	pga_10 (R/W)
	PGA coefficients 10		
R0x365E	15:0	0x0000	pga_11 (R/W)
	PGA coefficients 11		
R0x3660	15:0	0x0000	pga_12 (R/W)
	PGA coefficients 12		
R0x3662	15:0	0x0010	pga_13 (R/W)
	PGA coefficients 13		
R0x3664	15:0	0x0000	pga_14 (R/W)
	PGA coefficients 14		



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Table 24: SOC2 Registers (continued)

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x3666	15:0	0x0000	pga_15 (R/W)
	PGA coefficients 15		
R0x3668	15:0	0x0000	pga_16 (R/W)
	PGA coefficients 16		
R0x366A	15:0	0x0000	pga_17 (R/W)
	PGA coefficients 17		
R0x366C	15:0	0x0010	pga_18 (R/W)
	PGA coefficients 18		
R0x366E	15:0	0x0000	pga_19 (R/W)
	PGA coefficients 19		
R0x3670	15:0	0x0000	pga_20 (R/W)
	PGA coefficients 20		
R0x3672	15:0	0x0000	pga_21 (R/W)
	PGA coefficients 21		
R0x3674	15:0	0x0000	pga_22 (R/W)
	PGA coefficients 22		
R0x3676	15:0	0x0000	pga_23 (R/W)
	PGA coefficients 23		
R0x3678	15:0	0x0000	pga_24 (R/W)
	PGA coefficients 24		
R0x367A	15:0	0x0000	pga_25 (R/W)
	PGA coefficients 25		
R0x367C	15:0	0x0000	pga_26 (R/W)
	PGA coefficients 26		
R0x367E	15:0	0x0000	pga_27 (R/W)
	PGA coefficients 27		
R0x3680	15:0	0x0000	pga_28 (R/W)
	PGA coefficients 28		
R0x3682	15:0	0x0000	pga_29 (R/W)
	PGA coefficients 29		
R0x3684	15:0	0x0000	pga_30 (R/W)
	PGA coefficients 30		
R0x3686	15:0	0x0000	pga_31 (R/W)
	PGA coefficients 31		
R0x3688	15:0	0x0000	pga_32 (R/W)
	PGA coefficients 32		
R0x368A	15:0	0x0000	pga_33 (R/W)
	PGA coefficients 33		
R0x368C	15:0	0x0000	pga_34 (R/W)
	PGA coefficients 34		
R0x368E	15:0	0x0000	pga_35 (R/W)
	PGA coefficients 35		
R0x3690	15:0	0x0000	pga_36 (R/W)
	PGA coefficients 36		



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Table 24: SOC2 Registers (continued)

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x3692	15:0	0x0000	pga_37 (R/W)
	PGA coefficients 37		
R0x3694	15:0	0x0000	pga_38 (R/W)
	PGA coefficients 38		
R0x3696	15:0	0x0000	pga_39 (R/W)
	PGA coefficients 39		
R0x3698	15:0	0x0000	pga_40 (R/W)
	PGA coefficients 40		
R0x369A	15:0	0x0000	pga_41 (R/W)
	PGA coefficients 41		
R0x369C	15:0	0x0000	pga_42 (R/W)
	PGA coefficients 42		
R0x369E	15:0	0x0000	pga_43 (R/W)
	PGA coefficients 43		
R0x36A0	15:0	0x0000	pga_44 (R/W)
	PGA coefficients 44		
R0x36A2	15:0	0x0000	pga_45 (R/W)
	PGA coefficients 45		
R0x36A4	15:0	0x0000	pga_46 (R/W)
	PGA coefficients 46		
R0x36A6	15:0	0x0000	pga_47 (R/W)
	PGA coefficients 47		
R0x36A8	15:0	0x0000	pga_48 (R/W)
	PGA coefficients 48		
R0x36AA	15:0	0x0000	pga_49 (R/W)
	PGA coefficients 49		
R0x36AC	15:0	0x0000	pga_50 (R/W)
	PGA coefficients 50		
R0x36AE	15:0	0x0000	pga_51 (R/W)
	PGA coefficients 51		
R0x36B0	15:0	0x0000	pga_52 (R/W)
	PGA coefficients 52		
R0x36B2	15:0	0x0000	pga_53 (R/W)
	PGA coefficients 53		
R0x36B4	15:0	0x0000	pga_54 (R/W)
	PGA coefficients 54		
R0x36B6	15:0	0x0000	pga_55 (R/W)
	PGA coefficients 55		
R0x36B8	15:0	0x0000	pga_56 (R/W)
	PGA coefficients 56		
R0x36BA	15:0	0x0000	pga_57 (R/W)
	PGA coefficients 57		
R0x36BC	15:0	0x0000	pga_58 (R/W)
	PGA coefficients 58		



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Table 24: SOC2 Registers (continued)

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x36BE	15:0	0x0000	pga_59 (R/W)
	PGA coefficients 59		
R0x36C0	15:0	0x0000	pga_60 (R/W)
	PGA coefficients 60		
R0x36C2	15:0	0x0000	pga_61 (R/W)
	PGA coefficients 61		
R0x36C4	15:0	0x0000	pga_62 (R/W)
	PGA coefficients 62		
R0x36C6	15:0	0x0000	pga_63 (R/W)
	PGA coefficients 63		
R0x36C8	15:0	0x0000	pga_64 (R/W)
	PGA coefficients 64		
R0x36CA	15:0	0x0000	pga_65 (R/W)
	PGA coefficients 65		
R0x36CC	15:0	0x0000	pga_66 (R/W)
	PGA coefficients 66		
R0x36CE	15:0	0x0000	pga_67 (R/W)
	PGA coefficients 67		
R0x36D0	15:0	0x0000	pga_68 (R/W)
	PGA coefficients 68		
R0x36D2	15:0	0x0000	pga_69 (R/W)
	PGA coefficients 69		
R0x36D4	15:0	0x0000	pga_70 (R/W)
	PGA coefficients 70		
R0x36D6	15:0	0x0000	pga_71 (R/W)
	PGA coefficients 71		
R0x36D8	15:0	0x0000	pga_72 (R/W)
	PGA coefficients 72		
R0x36DA	15:0	0x0000	pga_73 (R/W)
	PGA coefficients 73		
R0x36DC	15:0	0x0000	pga_74 (R/W)
	PGA coefficients 74		
R0x36DE	15:0	0x0000	pga_75 (R/W)
	PGA coefficients 75		
R0x36E0	15:0	0x0000	pga_76 (R/W)
	PGA coefficients 76		
R0x36E2	15:0	0x0000	pga_77 (R/W)
	PGA coefficients 77		
R0x36E4	15:0	0x0000	pga_78 (R/W)
	PGA coefficients 78		
R0x36E6	15:0	0x0000	pga_79 (R/W)
	PGA coefficients 79		
R0x36E8	15:0	0x0000	pga_80 (R/W)
	PGA coefficients 80		



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Table 24: SOC2 Registers (continued)

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x36EA	15:0	0x0000	pga_81 (R/W)
	PGA coefficients 81		
R0x36EC	15:0	0x0000	pga_82 (R/W)
	PGA coefficients 82		
R0x36EE	15:0	0x0000	pga_83 (R/W)
	PGA coefficients 83		
R0x36F0	15:0	0x0000	pga_84 (R/W)
	PGA coefficients 84		
R0x36F2	15:0	0x0000	pga_85 (R/W)
	PGA coefficients 85		
R0x36F4	15:0	0x0000	pga_86 (R/W)
	PGA coefficients 86		
R0x36F6	15:0	0x0000	pga_87 (R/W)
	PGA coefficients 87		
R0x36F8	15:0	0x0000	pga_88 (R/W)
	PGA coefficients 88		
R0x36FA	15:0	0x0000	pga_89 (R/W)
	PGA coefficients 89		
R0x36FC	15:0	0x0000	pga_90 (R/W)
	PGA coefficients 90		
R0x36FE	15:0	0x0000	pga_91 (R/W)
	PGA coefficients 91		
R0x3700	15:0	0x0000	pga_92 (R/W)
	PGA coefficients 92		
R0x3702	15:0	0x0000	pga_93 (R/W)
	PGA coefficients 93		
R0x3704	15:0	0x0000	pga_94 (R/W)
	PGA coefficients 94		
R0x3706	15:0	0x0000	pga_95 (R/W)
	PGA coefficients 95		
R0x3708	15:0	0x0000	pga_96 (R/W)
	PGA coefficients 96		
R0x370A	15:0	0x0000	pga_97 (R/W)
	PGA coefficients 97		
R0x370C	15:0	0x0000	pga_98 (R/W)
	PGA coefficients 98		
R0x370E	15:0	0x0000	pga_99 (R/W)
	PGA coefficients 99		
R0x3710	15:0	0x0000	pga_100 (R/W)
	PGA coefficients 100		
R0x3712	15:0	0x0000	pga_101 (R/W)
	PGA coefficients 101		
R0x3714	15:0	0x0000	pga_102 (R/W)
	PGA coefficients 102		



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Table 25: SYSTL Registers

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x0000	15:0	0x2480	k24a_chip_id (RO)
	Model ID. Read-only.		
R0x0010	15:0	0x0180	pll_dividers (R/W)
	15:14	X	Reserved
	13:8	0x0001	n PLL N-Divider value
	7:0	0x0080	m PLL M-Divider value
	PLL dividers		
R0x0012	15:0	0x0000	pll_p_dividers (R/W)
	15:14	X	Reserved
	13:12	0x0000	word_clock_divider Division ratio of word_clock/word_clockn from bit_clock
	11:8	0x0000	p3 PLL system clock divider ratio P3
	7:4	0x0000	p2 PLL system clock divider ratio P2
	3:0	0x0000	p1 PLL system clock divider ratio P1
	PLL P dividers		



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Table 25: SYSTL Registers (continued)

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x0014	15:0	0x2145	pll_control (R/W)
	15	RO	pll_lock PLL lock status. This bit indicates when the PLL has acquired lock. After the PLL is enabled and the internal counters are reset, this bit becomes high when PLL lock is detected. 0: PLL lock is not detected. 1: PLL lock is detected.
	14	X	Reserved
	13	0x0001	clockin_hyst_en Hysteresis on EXTCLK enable 0: Hysteresis on EXTCLK is not enabled. 1: Hysteresis on EXTCLK is enabled.
	12:11	0x0000	Reserved
	10	0x0000	test_bypass Bypass the output of the VCO during the PLL initialization process. This bit should be assert when the PLL is enabled at the start of the PLL initialization. 0: VCO output is not bypassed. 1: VCO output is bypassed.
	9	0x0000	Reserved
	8	0x0001	reset_cntr Counters reset. This bit is used to reset the counters internally for the PLL to run after initialization. After the PLL is enabled (0x14[1]), reset the counters so the PLL will achieve lock before it is used. 0: Do not reset internal counters. 1: Reset internal counters.
	7:4	0x0004	Reserved
	3:2	0x0001	Reserved
	1	0x0000	pll_enable PLL enable. This bit enables the PLL, and should be enabled after the proper PLL dividers (0x10) and PLL System Clock Divider Ratio (0x12) have been initialized. 0: PLL is not enabled. 1: PLL is enabled.
	0	0x0001	pll_bypass Asserted to bypass the PLL. This should be asserted during PLL initialization. 0: PLL is not bypassed. 1: PLL is bypassed.
	PLL control		



MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Register Description

Table 25: SYSTL Registers (continued)

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x0016	15:0	0x0000	clocks_control (R/W)
	15	0x0000	Reserved
	14	0x0000	invert_output_clock PIXCLK invert. Invert the polarity of the pixel clock (PIXCLK). 0: PIXCLK is not inverted. 1: PIXCLK is inverted.
	13:10	0x0000	Reserved
	9	0x0000	clk_clkin_en Enable external clock signal 0: EXTCLK input is disabled. 1: EXTCLK input is enabled.
	8:0	0x0000	Reserved
	Clocks control		
R0x0018	15:0	0x4029	standby_control_and_status (R/W)
	15	RO	Reserved
	14	0x0001	standby_done Standby state indicator When soft standby state is initiated, this bit is asserted when standby state is achieved. 0: SOC is not in standby state. 1: SOC is in standby state.
	13:4	X	Reserved
	3	0x0001	en_xirq Standby control enable. This bit is used to disable the standby function (both STANDBY signal and the two-wire serial interface) of the sensor. 0: Standby control is not enabled. 1: Standby control is enabled.
	2:1	0x0000	Reserved
	0	0x0001	standby_i2c Soft standby bit This bit is set by default upon power-up, so the sensor starts up in soft standby mode. 0: SOC is not in soft standby. 1: SOC is in soft standby.
	Standby control and status		



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Table 25: SYSTL Registers (continued)

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x001A	15:0	0x0010	reset_and_misc_control (R/W)
	15:10	X	Reserved
	9	0x0000	parallel_enable Parallel output enable 0: Parallel output is disabled. 1: Parallel output is disabled.
	8	0x0000	oe_n_enable GPIO[2] enable. GPIO[2] can be configured as an output enable signal. 0: GPIO[2] does not function as OE. 1: GPIO[2] functions as OE.
	7:6	X	Reserved
	5	0x0000	stop_clkin_ip EXTCLK disable in standby. This gates the EXTCLK pin for the SOC during standby. 0: EXTCLK is not gated off during standby. 1: EXTCLK is gated off during standby.
	4	0x0001	ip_pd_en GPIO input signal power-down enable. When any GPIO signals are configured as inputs, this bit controls whether or not the input signal remains on during standby. 0: GPIO signals remains power-up during standby. 1: GPIO signals are turned off during standby.
	3:2	0x0000	Reserved
	1	0x0000	mipi_tx_reset MIPI Transmitter Reset 0: MIPI transmitter is not in reset. 1: MIPI transmitter is in reset.
	0	0x0000	reset_soc_i2c Soft reset. This reset is achieved by the serial two-wire interface, and its effect is the same as the RESET_BAR. 0: SOC is not in reset. 1: SOC is in reset.
	Reset and control		
R0x001C	15:0	0x0000	mcu_boot_mode (R/W)
	15:1	RO	Reserved
	0	0x0000	mcu_soft_rst_i2c This bit causes a soft reset. All sensor core registers and variables will be reset to their default states, programmed automatically by firmware. Some of the SOC registers will also be programmed by firmware during the soft reset. 0: MCU is not in reset 1: MCU is in reset
	MCU boot mode		



MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Register Description

Table 25: SYSTL Registers (continued)

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x001E	15:0	0x0400	signal_slew (R/W)
	15:11	X	Reserved
	10:8	0x0004	slew_pxlclk PIXCLK slew rate control
	7	X	Reserved
	6:4	0x0000	slew_gpio GPIO slew rate control
	3	X	Reserved
	2:0	0x0000	slew_io Parallel data output slew rate control
	Signal slew control		
R0x0022	15:0	0x0438	vdd_dis_counter (R/W)
	Internal standby delay. This bit controls the delay when STANDBY assertion takes effect. Default delay is 20 μ s when EXTCLK is 54 MHz. (Delay = 20 μ s = 0x0438/54 MHz.)		
R0x0024	15:0	0x0000	gpi_status (RO)
	The status of the GPIO signals are sampled at power up and the states of the signals are stored in this register. This is an alternate way for storing module ID.		
R0x0028	15:0	0x0001	hard_standby_sel (R/W)
	When the STANDBY signal is asserted, the sensor can enter one of two standby modes. The first mode maintains the registers and current sensor configuration. The second mode clears all the state information, restoring the sensor to its default state when the sensor comes out of standby. The latter state (no registers or configuration is saved) results in the lowest power consumption. This signal (hard_standby_sel) determines which state the sensor enters upon a STANDBY assertion: 0: The registers and state variables are retained when STANDBY is asserted. 1: The registers and state variables are not retained when STANDBY is asserted; resulting in the lowest power consumption.		

Table 26: XDMA Registers

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x098C	15:0	0x0000	mcu_variable_address (R/W)
	15	0x0000	access_8bit_mode 8-bit /16-bit Access Select 0: 16-bit access 1: 8-bit access
	14:13	0x0000	logical_select Select Logical Access Bits 14:13 of address for physical access; R0x98C[14:13] = 01 select logical access.
	12:8	0x0000	logical_id Driver ID Bits 12:8 of address for physical access; driver ID for logical access.
	7:0	0x0000	logical_offset Driver variable Bits 7:0 of address for physical access; driver variable offset for logical access.
	mcu variable address		



MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Register Description

Table 26: XDMA Registers (continued)

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x0990	15:0	0x0000	mcu_variable_data0 (R/W)
	To read a variable from MCU memory, set address in R0x098C and read data from R0x0990. To write to a variable, set address in R0x098C and write to R0x0990. 16-bit and 8-bit variables can be accessed, see R0x098C. When reading an 8-bit variable (R0x098C[15] = 1), R0x0990 [15:8] are set to 0. When writing to an 8-bit variable, R0x0990 [15:8] are ignored.		
R0x0992	15:0	0x0000	mcu_variable_data1 (R/W)
	Use these registers to read or write up to 16 bytes of variable data using the two-wire serial interface burst access mode. The variables must have consecutive addresses.		
R0x0994	15:0	0x0000	mcu_variable_data2 (R/W)
	Use these registers to read or write up to 16 bytes of variable data using the two-wire serial interface burst access mode. The variables must have consecutive addresses.		
R0x0996	15:0	0x0000	mcu_variable_data3 (R/W)
	Use these registers to read or write up to 16 bytes of variable data using the two-wire serial interface burst access mode. The variables must have consecutive addresses.		
R0x0998	15:0	0x0000	mcu_variable_data4 (R/W)
	Use these registers to read or write up to 16 bytes of variable data using the two-wire serial interface burst access mode. The variables must have consecutive addresses.		
R0x099A	15:0	0x0000	mcu_variable_data5 (R/W)
	Use these registers to read or write up to 16 bytes of variable data using the two-wire serial interface burst access mode. The variables must have consecutive addresses.		
R0x099C	15:0	0x0000	mcu_variable_data6 (R/W)
	Use these registers to read or write up to 16 bytes of variable data using the two-wire serial interface burst access mode. The variables must have consecutive addresses.		
R0x099E	15:0	0x0000	mcu_variable_data7 (R/W)
	Use these registers to read or write up to 16 bytes of variable data using the two-wire serial interface burst access mode. The variables must have consecutive addresses.		



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Table 27: Driver ID = 1: Sequencer Variables
R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x0002	15:0	0x000F	seq_mode (R/W)
	15:4	X	Reserved
	3	0x0001	seq_mode_hg Statistics engine 0: Statistics engine is disabled. 1: Statistics engine is enabled.
	2	0x0001	seq_mode_awb Auto white balance driver 0: Auto white balance is disabled. 1: Auto white balance is enabled.
	1	0x0001	seq_mode_fd Flicker detection driver 0: Flicker detection is disabled. 1: Flicker detection is enabled.
	0	0x0001	seq_mode_ae Auto exposure driver 0: Auto exposure is disabled. 1: Auto exposure is enabled.
	Set of 1-bit switches enabling various drivers and outdoor white balance option. Writing "1" to a particular switch enables the corresponding driver or option. Changes take effect immediately.		
R0x0003	15:0	0x0000	seq_cmd (R/W)
	Command or program to execute: 0x0000: Run 0x0001: Go to preview mode 0x0002: Go to capture mode 0x0003: Reserved 0x0004: Reserved 0x0005: Refresh 0x0006: Refresh mode Writing a positive value to this variable commands the sequencer to execute the corresponding program. The sequencer resets cmd to 0, executes the program, and then resumes running in its previous state. Changes take effect immediately. It is recommended that refresh and refresh mode commands always be run together, and that refresh mode should be issued before the refresh command.		
R0x0004	15:0	0x0003	seq_state (R/W)
	Current state of sequencer 0x0000: Reserved 0x0001: Reserved 0x0002: Enter preview 0x0003: Preview 0x0004: Leave preview 0x0005: Reserved 0x0006: Enter capture 0x0007: Capture 0x0008: Reserved 0x0009: Reserved		



MT9M113: 1/6-Inch 1.3Mp SOC Digital Image Sensor Register Description

Table 27: Driver ID = 1: Sequencer Variables (continued)

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x0006	15:0	0x0000	seq_flash_type (R/W)
	15:8	X	Reserved
	7	0x0000	seq_flash_type_flash_on_lock Set the flash to LOCK mode 0: Normal 1: LOCK mode
	6:0	0x0000	seq_flash_type_flash_type 0x0000: None 0x0001: LED 0x0002: Xenon 0x0003: Xenon burst
	Type of flash to be used		
R0x0009	15:0	0x0020	seq_ae_fast_buff (R/W)
	Weighting factor determining to what extent AE driver averages luma over time. Setting of 0x0020 results in the fastest AE speed while setting of 0x0001 results in the slowest AE speed.		
R0x000A	15:0	0x0001	seq_ae_fast_step (R/W)
	Number of steps in which AE driver is expected to reach target brightness. The smaller the number, the faster the AE reaches the target brightness. The bigger the number, the smoother the AE transitions.		
R0x000B	15:0	0x0008	seq_awb_cont_buff (R/W)
	Weighting factor determining to what extent AWB driver averages digital gains over time. Setting of 0x0020 results in the fastest AWB speed while setting of 0x0001 results in the slowest AWB speed.		
R0x000C	15:0	0x0002	seq_awb_cont_step (R/W)
	Number of steps in which AWB driver is expected to reach target color balance. The smaller the number, the faster the AWB reaches the target color balance. The bigger the number, the smoother the AWB transitions.		
R0x0011	15:0	0x00A8	seq_options (R/W)
	15:5	X	Reserved
	4	0x0000	seq_crop_win_awb Use crop window for AWB statistics.
	3	0x0001	seq_crop_win_ae Use crop window for AE statistics.
	2:0	0x0000	Reserved
	Sequence options		
R0x0013	15:0	0x0000	seq_flash_th (R/W)
	Exposure value (EV) threshold. If current EV is below this threshold, flash will be used in autoevaluate flash mode. Changes take effect immediately.		



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Table 27: Driver ID = 1: Sequencer Variables (continued)

R/W (Read or Write) bit; RO (Read Only) bit

Reg. #	Bits	Default	Name
R0x0015	15:0	0x0000	seq_cap_mode (R/W)
	15:7	X	Reserved
	6	0x0000	seq_cap_mode_video_fd_on Flicker Detection (FD) in capture and video mode 0: FD is disabled in capture and video mode. 1: FD is enabled in capture and video mode.
	5	0x0000	seq_cap_mode_video_awb_hg_on Auto White Balance (AWB) in capture and video mode. 0: AWB is disabled in capture and video mode. 1: AWB is enabled in capture and video mode.
	4	0x0000	seq_cap_mode_video_ae_hg_on Auto Exposure (AE) in capture and video mode 0: AE is disabled in capture and video mode. 1: AE is enabled in capture and video mode.
	3	X	Reserved
	2	0x0000	seq_cap_mode_flash_off_last_frame Turn flash off before last frame in capture state.
	1	0x0000	seq_cap_mode_video Video/Capture mode. When configured in capture mode, the sequencer returns to preview mode after the number of frames to be capture is completed. When configured in video mode, the sequencer does not return to preview mode until preview mode is selected for switch. 0: Capture mode 1: Video mode
	0	0x0000	seq_cap_mode_xenon_flash Capture with xenon flash 0: Taking snapshot does not enable xenon flash. 1: Taking snapshot enables xenon flash.
	Capture mode parameters. Take effect in capture state only.		
R0x0016	15:0	0x0003	seq_cap_num_frames (R/W)
	Number of frames captured in still capture mode. Changes take effect in capture mode only.		
R0x0017	15:0	0x0000	seq_preview_0_ae (R/W)
	Auto exposure configuration in preview-enter state 0x0000: Off 0x0001: Dynamic range tracking 0x0002: Brightness tracking 0x0003: Rules only (Ignore clipping)		
R0x0018	15:0	0x0000	seq_preview_0_fd (R/W)
	Flicker detection configuration in preview-enter state 0: Off 1: On 2: Manual		
R0x0019	15:0	0x0000	seq_preview_0_awb (R/W)
	Auto white balance configuration in preview-enter state 0: Off 1: On		